Does Ticket Price Influence Super Bowl Attendance?

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Data Background

- **Objective**: The goal of this project is to build a regression model that predicts Super Bowl attendance on a combination of team performance, ticket price, entertainment factors, venue characteristics, attendance, and host-city and state.
    - The aim is to identify which factors most strongly influence attendance levels across different Super Bowls and attempt to predict attendance for the 2026 Super Bowl

Data Retrieval

Variables:

- **Year** – Web scraped - NFL Playoff Pass
- **Teams (Winner and Loser)** – Web scraped - NFL Playoff Pass
- **Stadium** TopEnd Sports
- **Indoor/Outdoor** Wikipedia
- **Stadium Capacity** Wikipedia
- **Attendance** Statista
- **Average Ticket Price-** Yahoo Sports
- **City, State** TopEnd Sports
- **Type of Halftime Performance -** ESPN
- **Regular Season Win Percentage (Winner and Loser)** – Pro Football Reference

Data Characteristics

Field types

- **Numeric (continuous):** Attendance, Ticket Price, Winner RS Win %, Loser RS Win%, Year
- **Categorical (strings):** Team 1, Team 2, Stadium, City, State, Super_Bowl (Roman).
- **Booleans / binaries:** Indoor (0 = Outdoor, 1 = Indoor).
- **Ordinal category:** Type of Halftime Performance *****(1/2/3; see mapping below)

***** Halftime Performance (1/2/3):

1 = marching/themed or non-marquee act

2 = mid-level/mixed artists

3 = superstar headliner

Key Assumptions, Risks, and Current Decisions

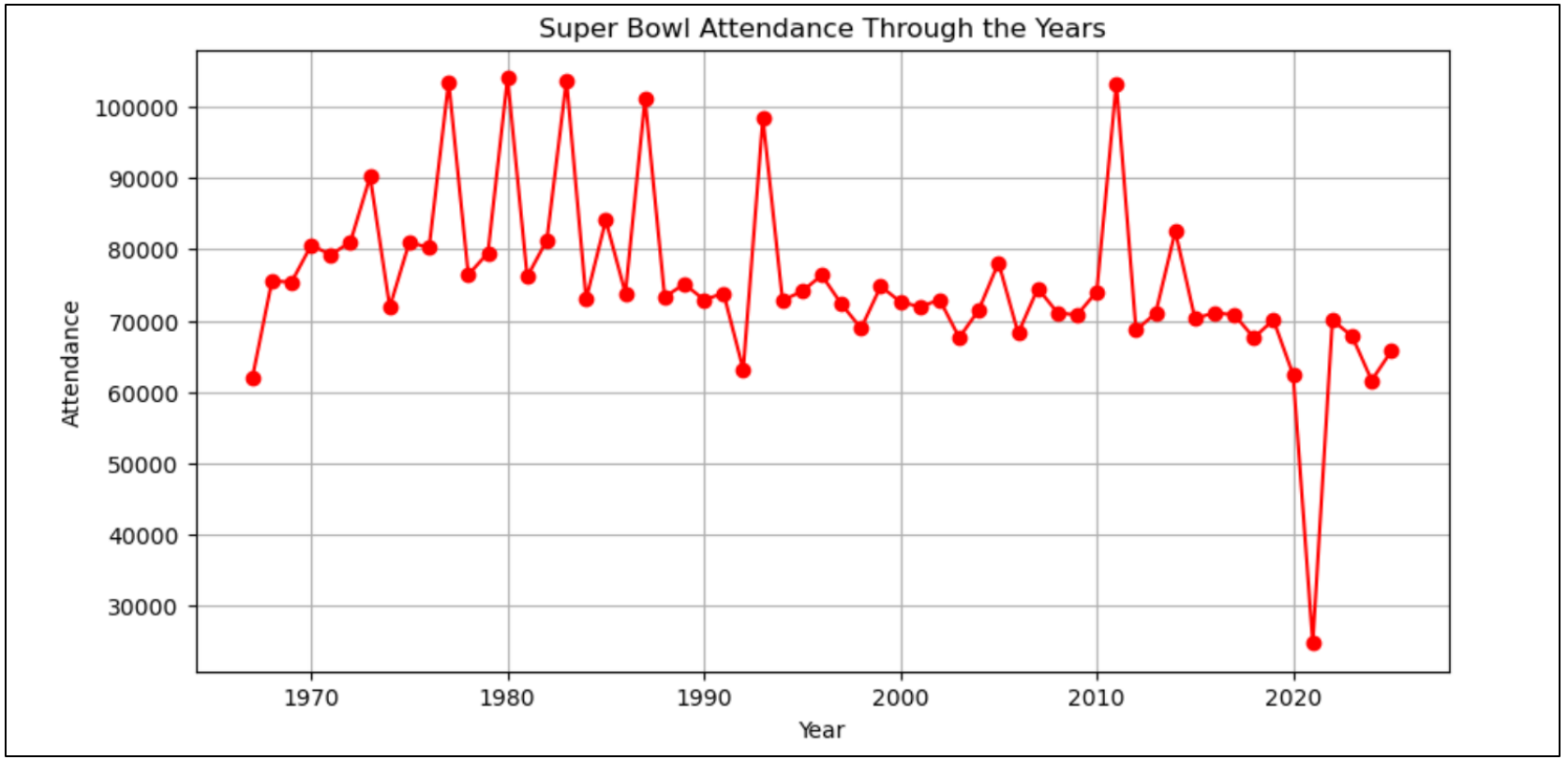
- COVID year (2020/2021) flagged for sensitivity analysis to handle extreme outlier effects
- Binary classification of indoor/outdoor
- 1,2,3 performance classification
- Regular season notes (players strikes, changes, etc.)
- Capacity
- Stadium/team/franchise evolution (stadiums being rebuilt/renamed, teams being renamed)

Data

Year	Team 1	Team 2	TicketPrice	Super_Bowl	Stadium	Indoor	City	State	Type_of_Halftime_Performance	Attendance	Winner_RS_WinPct	Loser_RS_WinPct	Capacity
2025	Philadelphia Eagles	Kansas City Chiefs	6304	LIX	Caesars Superdome	1	New Orleans	Louisiana	3	65719	0.824	0.882	73208
2024	Kansas City Chiefs	San Francisco 49ers	9400	LVIII	Allegiant Stadium	1	Paradise	Nevada	3	61629	0.647	0.706	65000
2023	Kansas City Chiefs	Philadelphia Eagles	9915	LVII	State Farm Stadium	1	Glendale	Arizona	3	67827	0.824	0.824	63400
2022	Los Angeles Rams	Cincinnati Bengals	7654	LVI	SoFi Stadium	1	Inglewood	California	2	70048	0.706	0.588	70240
2021	Tampa Bay Buccaneers	Kansas City Chiefs	6173	LV	Raymond James Stadium	0	Tampa	Florida	3	24835	0.688	0.875	69218

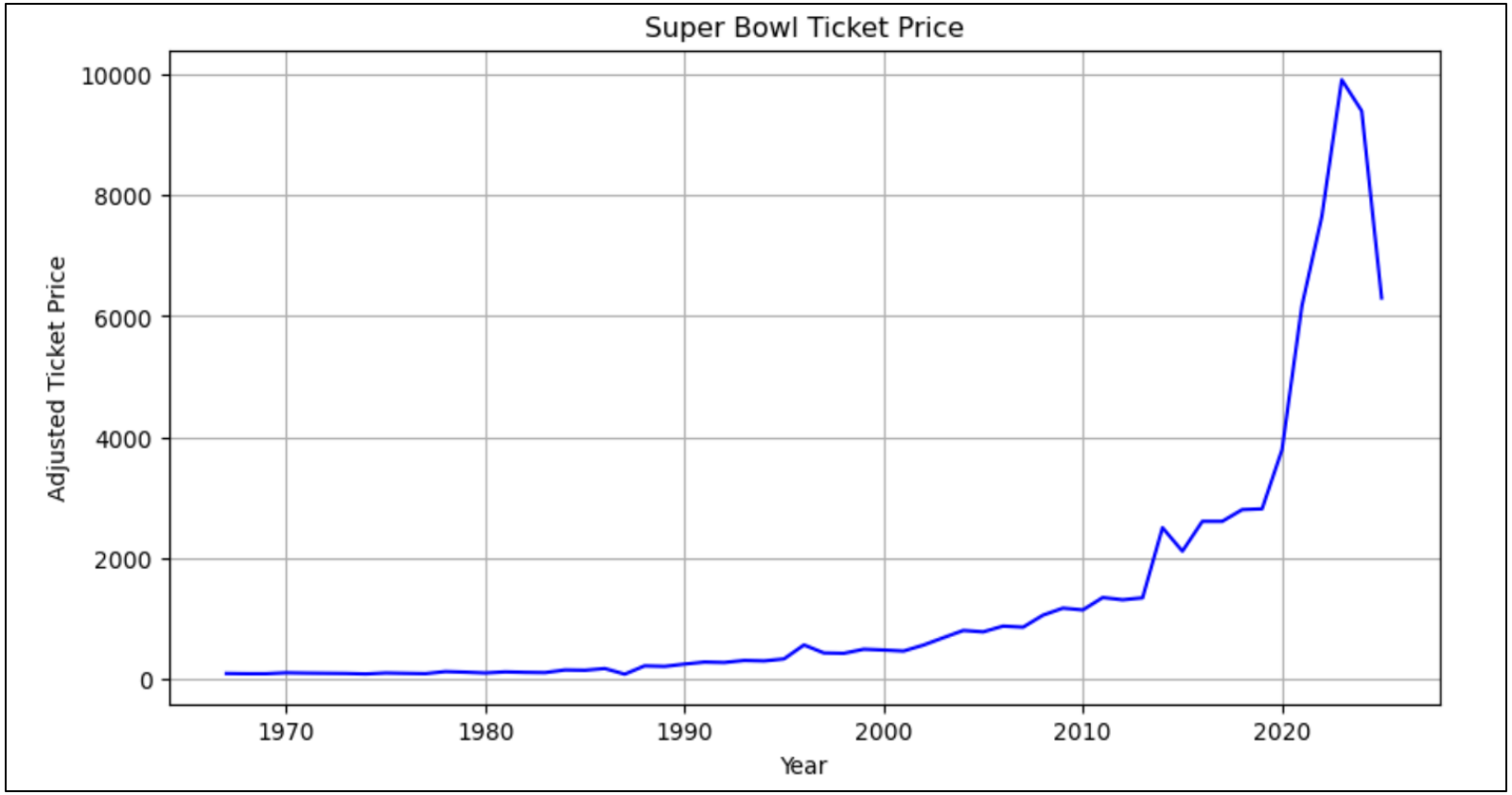
Note: Some stadiums have the capacity to expand their seating for special occasions. Thus, some Super Bowls' attendance with be larger than the Stadium capacity

EDA (Attendance)



- This graphs depicts the attendance to all the Super Bowls through the years
- The graph mainly fluctuates between 60,000 and around 100,000.
- See major fluctuations in the early years of the Super Bowl, but steadies starting around the late 1990s (see some outliers post-2010).

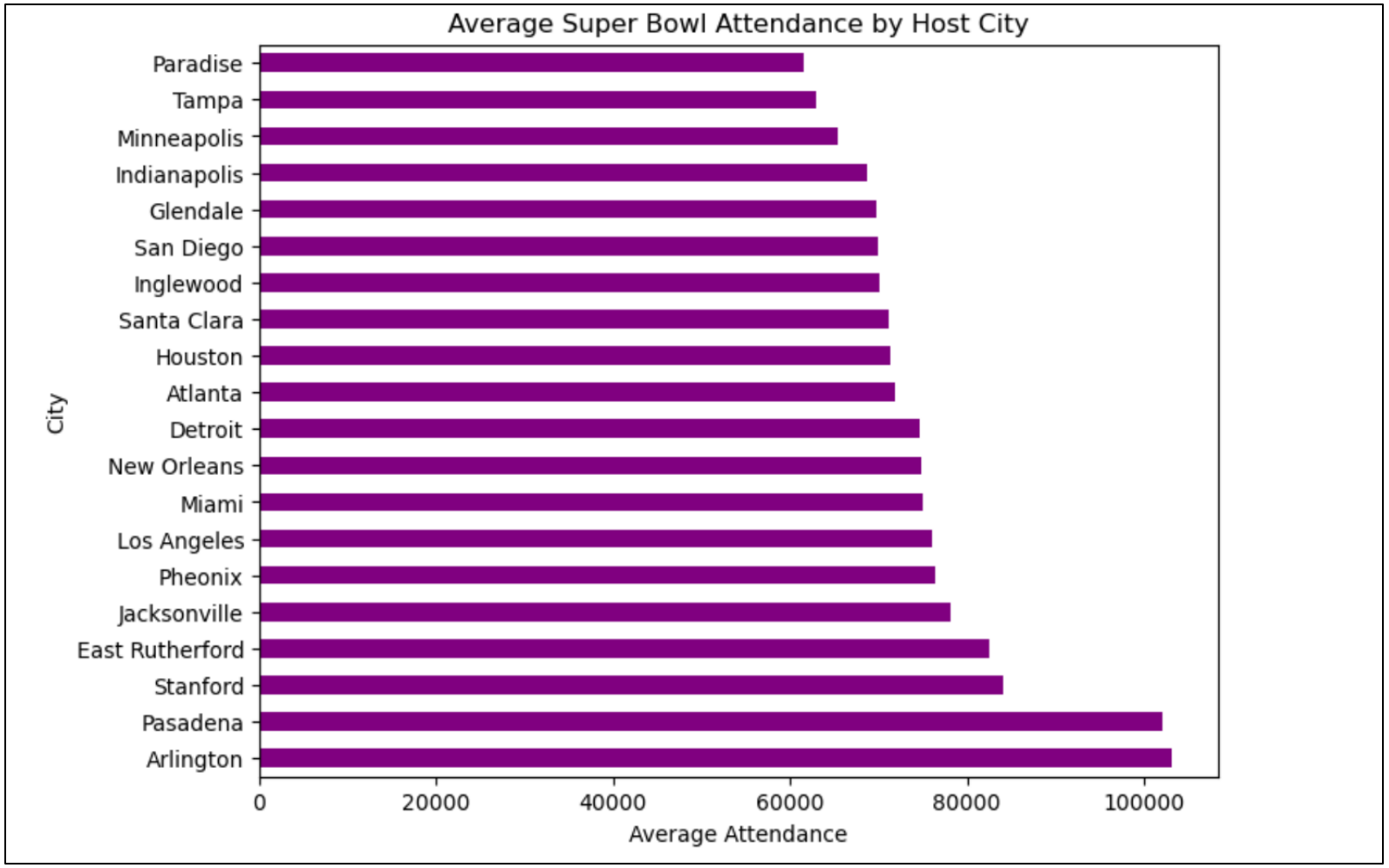
EDA (Ticket Price)



Mean: $1316
Standard Deviation: $2236

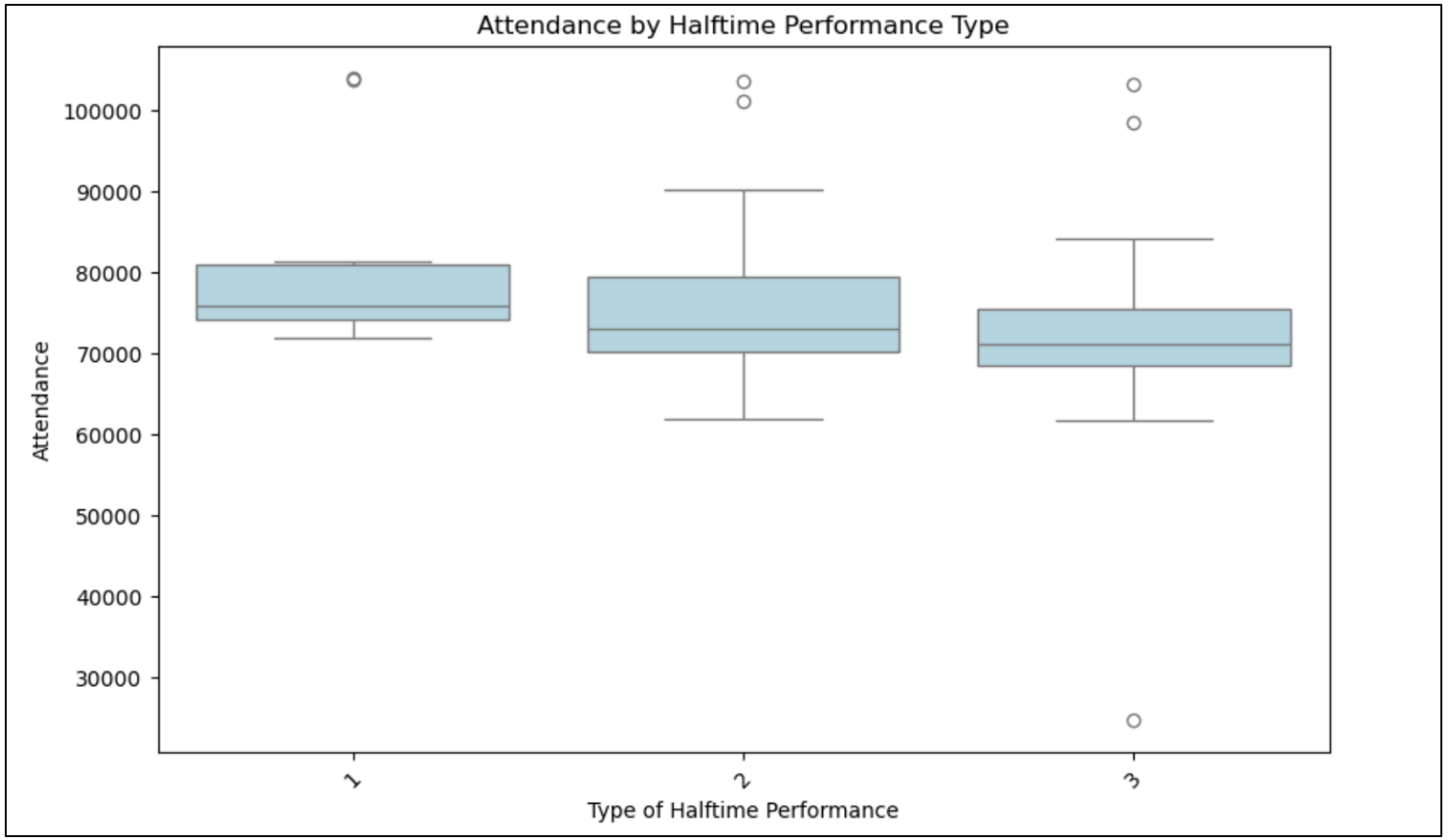
- The graph to the left depicts the ticket price for the Super Bowl through the years.
- Until around 2010, the ticket price remained steady.
- We see a huge spike around 2021and 2022 due to COVID-19 and the regulations surrounding the pandemic.
- Since COVID, the ticket prices have dropped.

EDA (Attendance by Host City)



- The chart shows the average attendance for the Super Bowls by host city.
- The attendance tends to be larger when the Super Bowl is hosted in Arlington, Pasadena and Stanford.

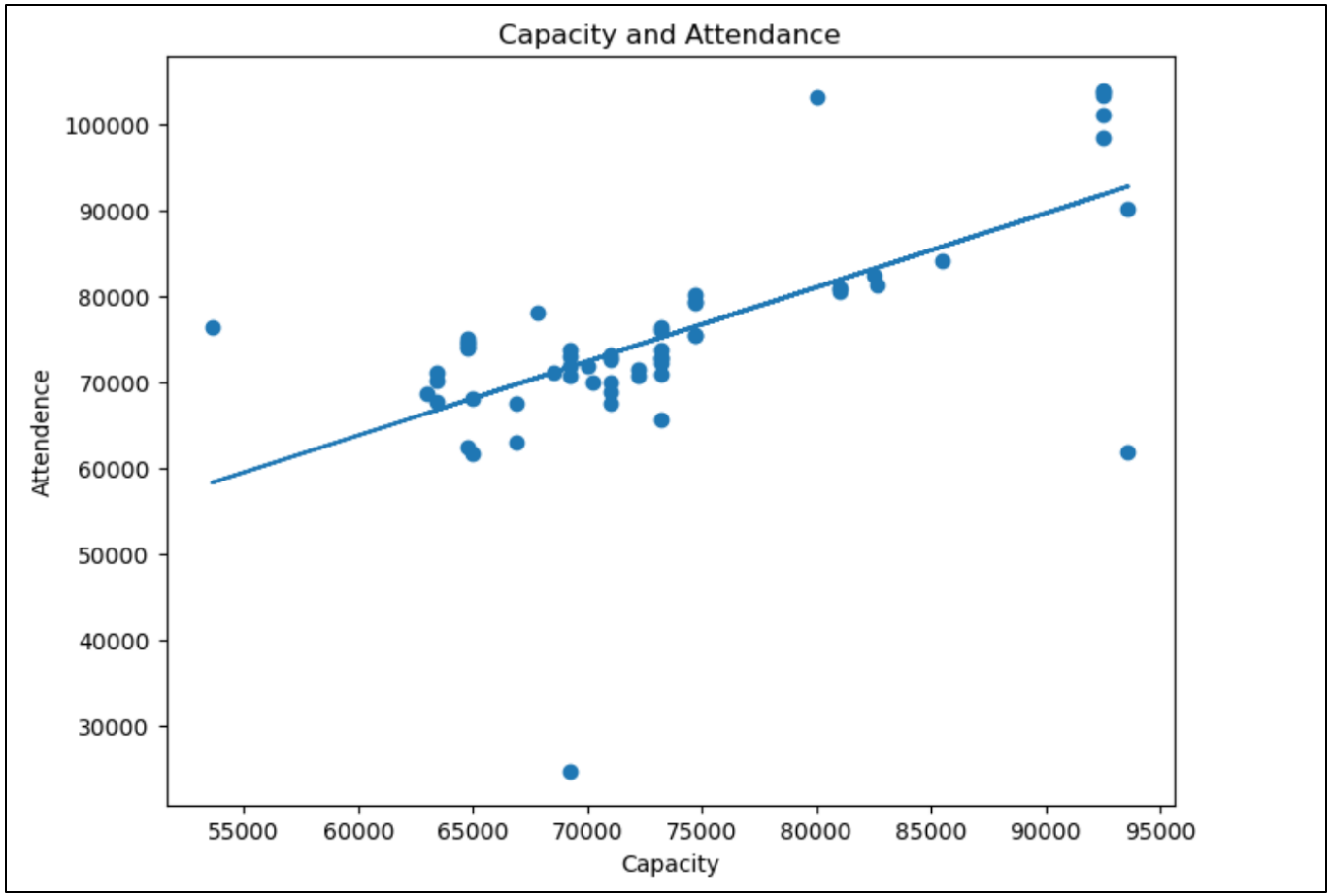
EDA (Attendance and Halftime)



1 = marching/themed or non-marquee act
2 = mid-level/mixed artists
3 = superstar headliner

- The box-and-whisker plot shows the that the median attendance is around the same for the 3 types of halftime performances.
- The box and whisker plot is the tightest for Type 1 indicating the attendance to be consistent.
- The plots are longer for types 2 and 3 showing more fluctuation.

EDA (Capacity and Attendance)



- Capacity and Attendance have a relatively strong correlation apart from several outliers.
- Many of the data points are consolidated between 65,000 and 75,000 regarding Capacity and 60,000 and 75,000 regarding Attendance.
- The major outliers occur due to the ability for the stadium to expand beyond its regular capacity

Why Multiple Linear Regression?

Our Goal: Predict Super Bowl Attendance (continuous numeric outcome)

With several potential drivers/predictors we can:

- Model attendance as a linear function of several predictors at once
- Control for other factors (test one while holding another constant)
- Get interpretable coefficients

With a small sample size, a linear model is less likely to overfit and is easier to explain results to stakeholders as opposed to more complex models

Our Model

- Used statsmodel formula API
- Dropped 2021 covid
- Fit an OLS (Ordinary Least Squares) model with predictors Capacity, Winner Regular Season Win %, Loser Regular Season Win %, Indoor, and Type of Halftime Performance
- Examined summary output for R^2, coefficients, and p-value

Categorical Halftime Variable

- Type_of_Halftime_Performance -> categorical variable with 3 types
- C(Type_of_Halftime_Performance) in the formula tells statsmodels to create dummy variables automatically
- Type 1 is treated as the baseline so the output only shows coefficients for T.2 and T.3
- Coefficients represent the average difference in attendance relative to Type 1, holding capacity, indoor, and win% constant
- This lets us formally test whether different halftime categories are associated with higher or lower attendance beyond stadium and team factors.

Model Results (1)

OLS Regression Results

Dep. Variable:	Attendance	R-squared:	0.595
Model:	OLS	Adj. R-squared:	0.547
Method:	Least Squares	F-statistic:	12.47
Date:	Mon, 08 Dec 2025	Prob (F-statistic):	1.35e-08
Time:	15:38:42	Log-Likelihood:	-592.37
No. Observations:	58	AIC:	1199.
Df Residuals:	51	BIC:	1213.
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	3.604e+04	1.44e+04	2.506	0.015	7165.447	6.49e+04
C(Type_of_Halftime_Performance)[T.2]	-4008.4368	2678.658	-1.496	0.141	-9386.068	1369.194
C(Type_of_Halftime_Performance)[T.3]	389.1713	2819.981	0.138	0.891	-5272.177	6050.520
Capacity	0.7977	0.118	6.738	0.000	0.560	1.035
Winner_RS_WinPct	-3132.4688	1.19e+04	-0.264	0.793	-2.69e+04	2.07e+04
Loser_RS_WinPct	-1.656e+04	1.07e+04	-1.548	0.128	-3.8e+04	4909.281
Indoor	-3674.9347	1986.090	-1.850	0.070	-7662.177	312.308

58 Super Bowls, 6 predictors
- explains ~55–60% of variation in attendance

Takeaway:
- Capacity is the clear driver of attendance
  - coef ≈ 0.798 ($p < 0.001$)
  - ~+800 attendees per +1,000 seats, holding other variables constant
- Other variables show weak evidence

Why run without Ticket Price

- Price is usually not a factor for people who are set on going to the Super Bowl (travel, accommodations, etc.), but could be a factor for locals

- Ticket price is often set based on expected demand, so it can partially “bake in” attendance expectations and make the model look stronger than it really is

Model Results (2)

OLS Regression Results

Dep. Variable:	Attendance	R-squared:	0.619
Model:	OLS	Adj. R-squared:	0.565
Method:	Least Squares	F-statistic:	11.59
Date:	Mon, 08 Dec 2025	Prob (F-statistic):	1.21e-08
Time:	15:37:29	Log-Likelihood:	-590.59
No. Observations:	58	AIC:	1197.
Df Residuals:	50	BIC:	1214.
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	3.925e+04	1.42e+04	2.764	0.008	1.07e+04	6.78e+04
C(Type_of_Halftime_Performance)[T.2]	-3572.3702	2635.143	-1.356	0.181	-8865.211	1720.471
C(Type_of_Halftime_Performance)[T.3]	1756.3968	2867.235	0.613	0.543	-4002.614	7515.408
Capacity	0.7694	0.117	6.574	0.000	0.534	1.005
Winner_RS_WinPct	-3031.1268	1.16e+04	-0.261	0.795	-2.64e+04	2.03e+04
Loser_RS_WinPct	-1.826e+04	1.05e+04	-1.736	0.089	-3.94e+04	2862.320
Indoor	-2565.8677	2042.966	-1.256	0.215	-6669.286	1537.550
TicketPrice	-0.8753	0.492	-1.777	0.082	-1.865	0.114

Compared to Model 1, fit improves slightly ----> R^2 +0.024 & Adj. R^2 +0.018

Takeaway:

- Higher ticket price is *associated* with *slightly* lower attendance after controlling for capacity, etc. (but it's not strong enough to call "significant") -> weak negative relationship
- Capacity still explains most of what we can predict about attendance
- Other factors still don't show clear independent effects in this model.

2026 Superbowl Attendance Prediction

Assumed Inputs (2026):

Stadium Capacity
- 68,500 (Levi's Stadium)

Indoor
- 0 (Outdoor)

Halftime Type
- 3 ("superstar performer")

Ticket Price
- Historical Average

Winner RS Win%
- Historical Average

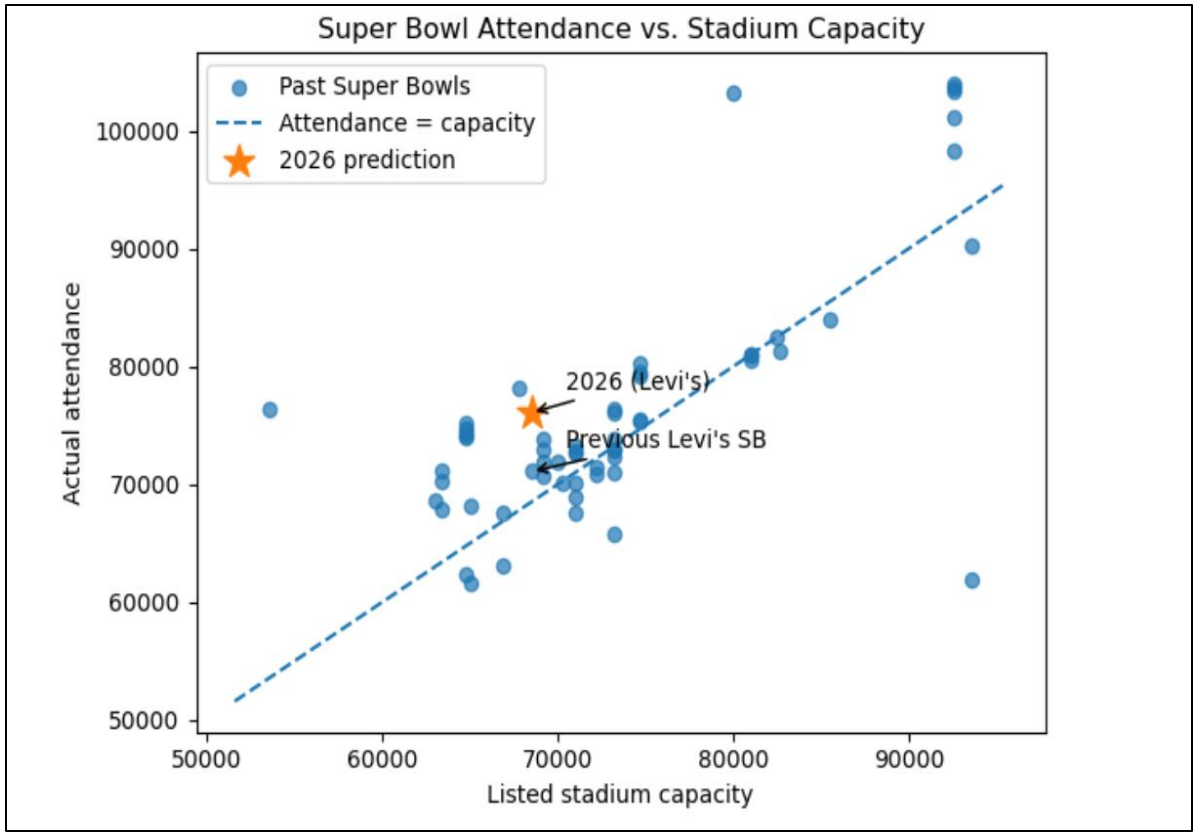
Loser RS Win%:
- Historical Average

Prediction:

- Predicted attendance: 76,079
- Expected mean (95% CI): 71,994 – 79,523



2026 Prediction



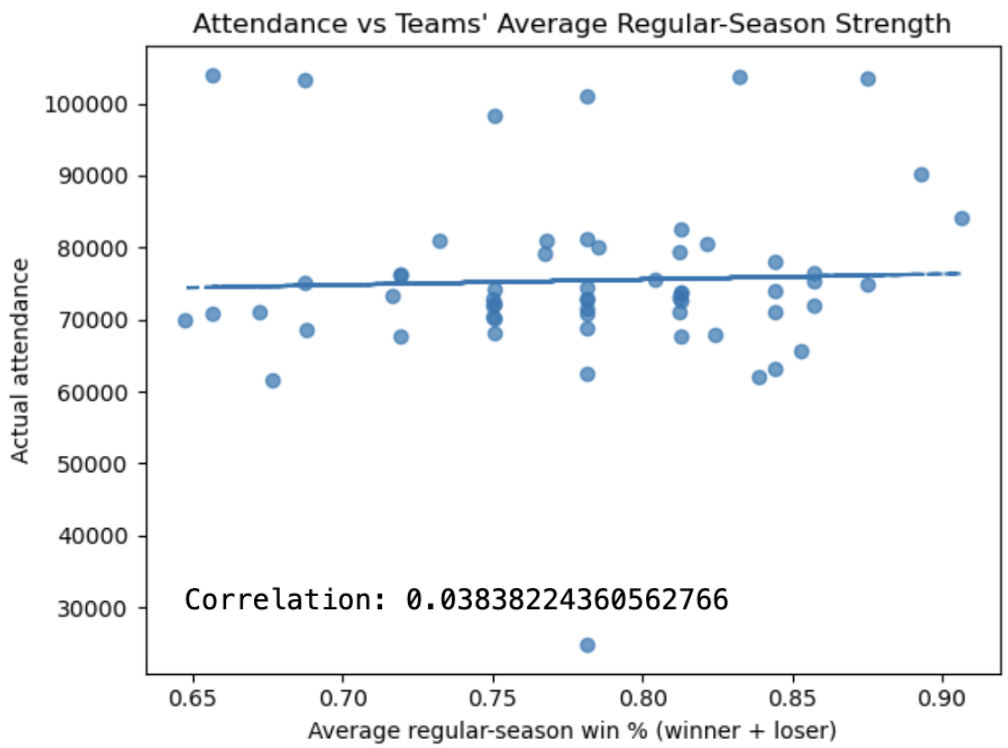
- Positive relationship between Capacity and Attendance (larger capacity is associated with higher attendance)
- Many games are played at or near listed capacity, indicated sell-outs or near sell-outs
- Some games fall noticeably below stadium capacity indicating that not every Super Bowl fully uses the stadium's potential
- Some games exceed listed capacity due to temporary seating, standing room sections, or alternate configurations for major events
- The orange star sits above the equality line, but in the same general region as other over-capacity games and close to the Levi's previous Super Bowl, suggesting the prediction is high, but still in line with historical patterns

Regular Season Win Percentage

Year	Avg_RS_WinPct	RS_WinPct_Gap	Attendance
2025	0.8530	0.058	65719
2024	0.6765	0.059	61629
2023	0.8240	0.000	67827
2022	0.6470	0.118	70048
2021	0.7815	0.187	24835

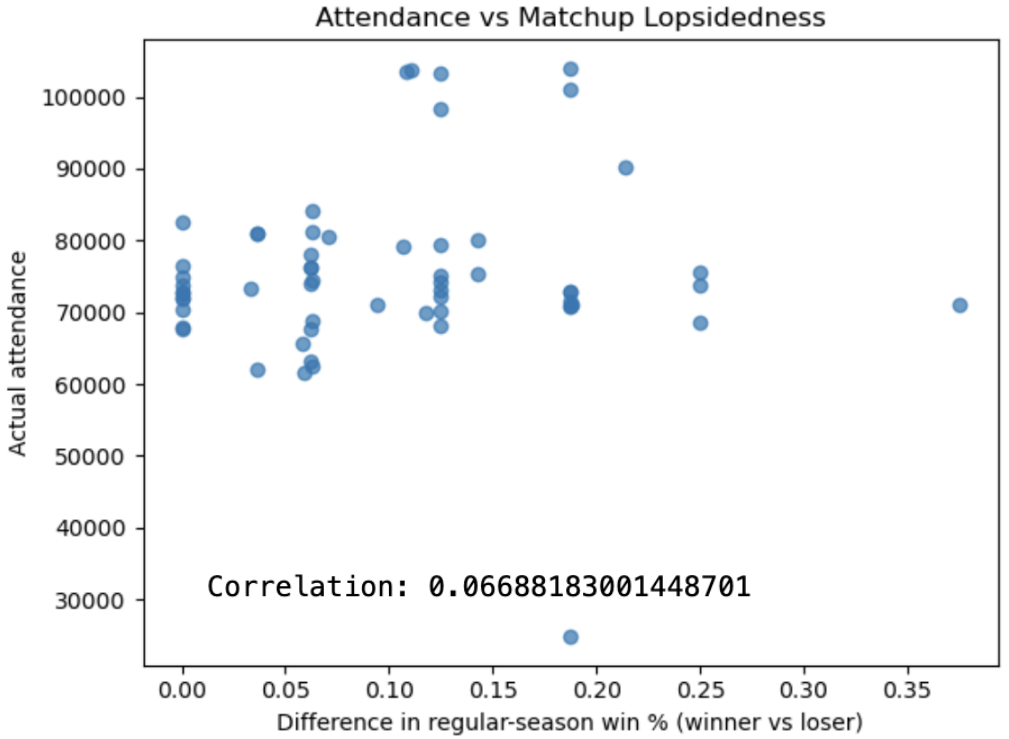
- Want to see if the rank of the team influences the attendance
- Initially wanted to use the league rank, but opted to use win percentage instead
- Average – used to check whether the matchups between stronger teams (higher average win percentage) draw more fans
- Gap – how far apart were the teams in standings

Regular Season Win Percentage



- Correlation ≈ 0.04 → basically zero.
- For any given average win % (0.65, 0.75, 0.85…), attendance is spread all over the place: mid-60Ks, 70Ks, even 100K+.
- Super Bowls with two really strong teams (high average win %) do not have systematically higher attendance than games with more average teams.

---> Matches regression results



- Correlation ≈ 0.07 → again very close to zero.
- Games with a small gap (very evenly matched) and games with a big gap (strong favorite vs underdog) are all over the place in attendance.
- There's one obvious low-attendance outlier, but otherwise no trend across the x-axis.
- Whether the matchup is even or lopsided doesn't show a clear relationship with attendance either.
- Blowout-looking matchups and tight matchups both draw similar crowds

Regular Season Win Percentage Conclusion

- Across all Super Bowls, attendance does not systematically vary with either average team strength or how lopsided the matchup is (correlations = 0.04 and 0.07)
- There are a few attendance outliers (one unusually small crowd, a few unusually large ones), which line up with unusual venue choices or configurations, but not with team strength
- This supports our regression results: once we control for stadium capacity, indoor/outdoor, and ticket price, who the teams are and how close the matchup looks on paper do not play a big role in how many people show up

Overall Conclusion

- Stadium Capacity is the main predictor of Superbowl Attendance
- Adding ticket price improved the model slightly but still had a weak negative relationship with attendance
- 2026 prediction ≈ 76, 079 people
- Key Limitations

  - small sample size (58 games after dropping 2021), outlier years (COVID), listed stadium capacity

  Big Picture: Stadium/venue characteristics more reliably predicts attendance; the matchup of teams and halftime show are much less predictive.

Implications for Stakeholders

- **Super Bowl Organizers**: Accurate prediction in Super Bowl attendance assists the NFL in optimization of fan experience, operations, and revenue
- **Host Stadium**: The stadium can properly staff and stock resources
- **Media**: Assists the media in planning coverage, allocate responsibilities, and enhance fan experience
- **Ticketing Platforms**: guides pricing and manages demand
- **Businesses associated with NFL and Super Bowl**: understand how to allocate resources and set prices

Ethical, Legal, Societal Implications

Societal

- Crowd Management and Safety: knowing the attendance helps mitigate risks and enhances fan enjoyment
- Urban Planning: the host city will be able to properly prepare for the wave of visiting fans and maximize city efficiency
- Cultural Experience: predictions will influence how stadiums and tailgating areas are designed affecting fan interactions and fan experience as a whole

Ethical

- Data Bias: our model uses historical data which does not always account for all possible inequities that occurred in the past such as underrepresentation or lower-income fans
- Ticket Accessibility: ticketing platforms and algorithms often maximize revenue

Legal

- Follow copyright and terms-of-use rules for any websites, APIs, or datasets used
- Future use of the model's outputs complies with consumer protection